Theorem 14: upper bound for the spectral distance

Correspondence between the paper and the Lean 4/Mathlib formalisation

Paper. J. Breitung, T. Gervens, M. Grohe. *Graph Similarity Meets Optimal Transport* (file `Graph_Similarity_Meets_Optimal_Transport.pdf` in the repository). Statement, equation, page and line numbers refer to this PDF; `docs/NUMBERING.md` lists the Lean files and declarations of all numbered statements.

The folder `GsimOT/Theorem15_SpectralUpperBound` formalises Theorem 14, its proof in App. F.2 with Lemmas 50–52, and the δ_2 part of Theorem 2. No cited results are assumed.

Checking. Run `bash build.sh` in the repository root (see `README.md`). The script builds the library and runs `scripts/CheckAxioms.lean`, which must report only the standard axioms `propext`, `Classical.choice`, `Quot.sound`: no `sorry` and no additional axioms.

Main statement in Lean. `GsimOT.theorem14`: for all graphs G, H of the same order, with $\delta = \delta_2^{\mathrm{OT}}(G,H)$,

$$\delta_2(G,H) \le \delta + 2\sqrt{2\delta}.$$

`GsimOT.theorem15_h`: $\delta_2(G,H) \le \delta + 4\sqrt{h\delta}$ with $h = \min\{\sqrt{p_G(1-p_G)}, \sqrt{p_H(1-p_H)}\}$ (end of the proof); `GsimOT.theorem15`: $\delta_2(G,H) \le \min\{1, \delta + 4\sqrt{h\delta}\} \le \min\{1, \delta + 2\sqrt{2\delta}\}$ (with $\delta_2 \le 1$); `GsimOT.theorem15_sqrt`: $\delta_2(G,H) \le (1+2\sqrt{2})\sqrt{\delta}$.

Correspondence

Paper	Page	Statement	Lean declaration(s) and location
Theorem 14	12	$\delta_2 \le \delta_2^{\mathrm{OT}} + 2\sqrt{2\delta_2^{\mathrm{OT}}}$	`GsimOT.theorem14` `Theorem15_SpectralUpperBound/Theorem15.lean:315`
end of proof of Thm. 14	50	$\delta_2 \le \gamma_Q + 4\sqrt{\gamma_Q h}$ with $\gamma_Q = \delta_2^{\mathrm{OT}}$ (l. 1573–1574)	`GsimOT.theorem15_h` `Theorem15_SpectralUpperBound/Theorem15.lean:291`

Thm. 14 and end of its proof	12, 50	$\delta_2 \le \min\{1, \delta_2^{\mathrm{OT}} + 4\sqrt{h\,\delta_2^{\mathrm{OT}}}\} \le \min\{1, \delta_2^{\mathrm{OT}} + 2\sqrt{2\delta_2^{\mathrm{OT}}}\}$: the bound of the proof combined with $\delta_2 \le 1$ ($\lVert\cdot\rVert_2 \le \lVert\cdot\rVert_F \le n$)	`GsimOT.theorem15`
			`Theorem15_SpectralUpperBound/Theorem15.lean:326`
			`GsimOT.deltaPQ_two_two_le_one`
			`Theorem15_SpectralUpperBound/Theorem15.lean:267`
			`GsimOT.Theorem15.distPQ_le_card`
			`Theorem15_SpectralUpperBound/Theorem15.lean:192`

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after Thm. 14; Thm. 2 (δ_2)	3, 12	$\delta_2^{\mathrm{OT}} \le 1$, hence $\delta_2 \le (1+2\sqrt{2})\sqrt{\delta_2^{\mathrm{OT}}}$ (l. 399)	`GsimOT.theorem15_sqrt`
			`Theorem15_SpectralUpperBound/Theorem15.lean:339`
			`GsimOT.deltaOTPQ_two_two_le_one`
			`Theorem15_SpectralUpperBound/Theorem15.lean:277`
			`GsimOT.opNorm_Bmat_le_sum`
			`Theorem15_SpectralUpperBound/Theorem15.lean:242`

App. F.2	47	Spectral norm: $\|\cdot\|_{2\to2}$ equals Mathlib's ℓ^2 operator norm; $\|K\|_2 \le \|K\|_F$; transposition, reindexing (l. 1452–1453)	`GsimOT.opNorm_two_two_eq`
			`Theorem15_SpectralUpperBound/Tools.lean:73`
			`GsimOT.l2_norm_le_frobenius`
			`Theorem15_SpectralUpperBound/Tools.lean:86`
			`GsimOT.l2_norm_transpose`
			`Theorem15_SpectralUpperBound/Tools.lean:118`
			`GsimOT.l2_norm_submatrix`
			`Theorem15_SpectralUpperBound/Tools.lean:136`

App. F.2	47, 50	$\|K\|_F^2 = \sum_r \|K\phi_r\|_2^2$ for an orthonormal basis (l. 1453; in the proof of Eq. (43), l. 1554–1556)	`GsimOT.Theorem15.frobenius_eq_sum_eigvec` `Theorem15_SpectralUpperBound/RandomBijection.lean:432`

Eq. (40)	47	$\gamma_Q(A',B')$: $n\gamma_Q = \|B_{2\to2}\|_2$ (kernel $\sqrt{Q}\,D\,\sqrt{Q}$); the bilinear bound	`GsimOT.Theorem15.discKernel`
			`Theorem15_SpectralUpperBound/RandomBijection.lean:54`
			`GsimOT.Theorem15.eq40_le`
			`Theorem15_SpectralUpperBound/RandomBijection.lean:95`
			`GsimOT.Theorem15.discKernel_shift`
			`Theorem15_SpectralUpperBound/Theorem15.lean:84`

Eq. (41)	47	$A - PBP^\top = (A - RAR^\top) + P(Q^\top AQ - B)P^\top$, $R = PQ^\top$	`GsimOT.Theorem15.eq41` `Theorem15_SpectralUpperBound/RandomBijection.lean:328`
Lemma 50	47	$n^{-1}\|Q^\top A'Q - B'\|_2 \le \gamma_Q(A',B')$ (via Lemma 19)	`GsimOT.Theorem15.lemma50` `Theorem15_SpectralUpperBound/RandomBijection.lean:168`

App. F.2	47	$\|R\|_2 \le 1$ for $R \in \mathrm{DS}(n)$; $\|P\|_2 \le 1$ (l. 1485–1486)	`GsimOT.l2_norm_le_one_of_stochastic`
			`Theorem15_SpectralUpperBound/Tools.lean:106`
			`GsimOT.Theorem15.norm_pmat_mul_transpose_le`
			`Theorem15_SpectralUpperBound/RandomBijection.lean:306`
			`GsimOT.Theorem15.norm_pmat_le`
			`Theorem15_SpectralUpperBound/RandomBijection.lean:295`

Paper	Page	Statement	Lean declaration(s) and location
Eq. (28)	47, 50	$X - RXR^\top = EX + R(EX)^\top$, $E = I - R$, for symmetric X	`GsimOT.Theorem15.eq28` `Theorem15_SpectralUpperBound/RandomBijection.lean:336`

Eq. (42)	48	$\frac{1}{n^2}\sum_{ij} Q_{ij}F_{ij}^2 \le \gamma_Q^2\|y\|^2$, $F_{ij} = x_i - r_j$	`GsimOT.Theorem15.Fmat`
			`Theorem15_SpectralUpperBound/RandomBijection.lean:204`
			`GsimOT.Theorem15.Fmat_eq`
			`Theorem15_SpectralUpperBound/RandomBijection.lean:208`
			`GsimOT.Theorem15.eq42`
			`Theorem15_SpectralUpperBound/RandomBijection.lean:215`

Lemma 51	48	Local agreement: $\frac{1}{n^2}\sum_{ij} Q_{ij}(x_i - (Q^\top x)_j)^2 \le \gamma_Q^2\|y\|^2$ for $x = A'y$	`GsimOT.Theorem15.lemma51`
			`Theorem15_SpectralUpperBound/RandomBijection.lean:247`
			`GsimOT.sum_weighted_sq_sub`
			`Theorem15_SpectralUpperBound/Tools.lean:148`

Lemma 52	48–49	$\mathbb{E}\|(I-R)x\|_2^2 = \sum_{ij} Q_{ij}(x_i - (Q^\top x)_j)^2$ for the random permutation of a Birkhoff decomposition	`GsimOT.Theorem15.lemma52`
			`Theorem15_SpectralUpperBound/RandomBijection.lean:361`
			`GsimOT.Theorem15.sum_birkhoff_eq`
			`Theorem15_SpectralUpperBound/RandomBijection.lean:346`
			`GsimOT.exists_birkhoff_decomposition`
			`Common/BirkhoffDecomposition.lean:42`

Proof of Thm. 14	49	Q optimal: $\gamma_Q = \delta_2^{\mathrm{OT}}$ (l. 1526)	`GsimOT.exists_dOT_eq`
			`Common/Attainment.lean:223`
			`GsimOT.deltaOTPQ_eq_dOT`
			`Common/OTDistances.lean:131`

Proof of Thm. 14	49	Centring: $p_G = 2\|E(G)\|/n^2$, $A' = A - p_G J$, $B' = B - p_G J$; $D_{A',B'} = D_{A,B}$, $A' - PB'P^\top = A - PBP^\top$, $\|A'\|_F^2 = n^2h^2$ (l. 1528–1532)	`GsimOT.edgeDensity`
			`Theorem15_SpectralUpperBound/Theorem15.lean:44`
			`GsimOT.edgeDensity_eq`
			`Theorem15_SpectralUpperBound/Theorem15.lean:49`
			`GsimOT.Theorem15.shiftAdj`
			`Theorem15_SpectralUpperBound/Theorem15.lean:67`
			`GsimOT.Theorem15.discKernel_shift`
			`Theorem15_SpectralUpperBound/Theorem15.lean:84`
			`GsimOT.Theorem15.aligned_shift`
			`Theorem15_SpectralUpperBound/Theorem15.lean:94`
			`GsimOT.Theorem15.sum_sq_shiftAdj`
			`Theorem15_SpectralUpperBound/Theorem15.lean:101`

Paper	Page	Statement	Lean declaration(s) and location
Proof of Thm. 14	49	Spectral truncation A'_t, I_t; $\|A' - A'_t\|_2 \le nt$, $\|I_t\| \le h^2/t^2$ (l. 1533–1544)	GsimOT.Theorem15.truncHigh Theorem15_SpectralUpperBound/RandomBijection.lean:469 GsimOT.Theorem15.truncLow Theorem15_SpectralUpperBound/RandomBijection.lean:473 GsimOT.Theorem15.ind Theorem15_SpectralUpperBound/RandomBijection.lean:460 GsimOT.Theorem15.norm_truncLow_le Theorem15_SpectralUpperBound/RandomBijection.lean:484 GsimOT.Theorem15.sq_mul_card_le Theorem15_SpectralUpperBound/RandomBijection.lean:513
Eq. (43)	49	$\mathbb{E}\|EA'_t\|_2^2 \le \mathbb{E}\|EA'_t\|_F^2 = \sum_{r\in I_t} \mathbb{E}\|EA'\phi_r\|_2^2 \le n^2\gamma_Q^2\|I_t\|$	GsimOT.Theorem15.eq43 Theorem15_SpectralUpperBound/RandomBijection.lean:550 GsimOT.Theorem15.frobenius_one_sub_R_truncHigh Theorem15_SpectralUpperBound/RandomBijection.lean:531 GsimOT.Theorem15.truncHigh_mulVec_eigvec Theorem15_SpectralUpperBound/RandomBijection.lean:507 GsimOT.Theorem15.errBij Theorem15_SpectralUpperBound/RandomBijection.lean:526
Proof of Thm. 14	50	Some P with $\|EA'_t\|_2 \le n\gamma_Q h/t$ (l. 1561–1562)	GsimOT.Theorem15.exists_one_sub_R_truncHigh_le Theorem15_SpectralUpperBound/RandomBijection.lean:578 GsimOT.Theorem15.exists_le_of_sum_mul_le Theorem15_SpectralUpperBound/RandomBijection.lean:369
chain (a), p. 50	50	$\|A' - RA'R^\top\|_2 \le 2nt + 2n\gamma_Q h/t$; with Eq. (41), Lemma 50: $\gamma_P \le \gamma_Q + 2t + 2\gamma_Q h/t$	GsimOT.Theorem15.exists_bij_norm_le' Theorem15_SpectralUpperBound/RandomBijection.lean:609 GsimOT.Theorem15.exists_bij_norm_le Theorem15_SpectralUpperBound/RandomBijection.lean:658 GsimOT.Theorem15.exists_aligned_le Theorem15_SpectralUpperBound/RandomBijection.lean:687 GsimOT.Theorem15.exists_aligned_le' Theorem15_SpectralUpperBound/RandomBijection.lean:713 GsimOT.Theorem15.deltaPQ_le_threshold_G Theorem15_SpectralUpperBound/Theorem15.lean:143 GsimOT.Theorem15.deltaPQ_le_threshold_H Theorem15_SpectralUpperBound/Theorem15.lean:159

Paper	Page	Statement	Lean declaration(s) and location
Proof of Thm. 14	50	Choice $t = \sqrt{\gamma_Q h}$ (or $t \to 0$); $h \le \frac{1}{2}$ (l. 1573–1575)	`GsimOT.Theorem15.le_of_forall_threshold` Theorem15_SpectralUpperBound/RandomBijection.lean:729 `GsimOT.Theorem15.deltaPQ_le_of_isDS` Theorem15_SpectralUpperBound/Theorem15.lean:176 `GsimOT.min_sqrt_density_le_half` Theorem15_SpectralUpperBound/Theorem15.lean:220 `GsimOT.four_sqrt_le` Theorem15_SpectralUpperBound/Theorem15.lean:230

Shared notions used (from `GsimOT/Common`)

Paper	Page	Statement	Lean declaration(s) and location
Sec. 1, App. A	2–3, 14	Couplings $\mathrm{DS}^*(V,W)$; x^a with $0^a := 0$ (convention $0^0 = 0$)	`GsimOT.IsCoupling` Common/OTDistances.lean:36 `GsimOT.powZ` Common/OTDistances.lean:42
Sec. 1, Eqs. (1), (2)	2–3	$B(G,H,Q)$ and $B_{p\to q}(G,H,Q)$	`GsimOT.B` Common/OTDistances.lean:58 `GsimOT.Bmat` Common/OTDistances.lean:70
Eq. (2), Sec. 2	3, 5	$\delta^{\mathrm{OT}}_{p\to q}$ (couplings) and $d^{\mathrm{OT}}_{p\to q}$ ($\mathrm{DS}(V,W)$), both infima	`GsimOT.deltaOTPQ` Common/OTDistances.lean:74 `GsimOT.dOT` Common/OTDistances.lean:79
Secs. 1–2, Apps. A–B	3, 5, 14–15	$\delta^{\mathrm{OT}}_{p\to q} = n^{-\kappa_{p\to q}} d^{\mathrm{OT}}_{p\to q}$ for graphs of the same order n ($d^{\mathrm{OT}}_{p\to q}$ is the unnormalised version, $s(n) = n^{\kappa_{p\to q}}$), via $\mathrm{DS}(V,W) = n\,\mathrm{DS}^*(V,W)$ and the homogeneity of $B_{p\to q}$ and of $\|\cdot\|_{p\to q}$	`GsimOT.deltaOTPQ_eq_dOT` Common/OTDistances.lean:131 `GsimOT.Bmat_smul` Common/OTDistances.lean:117 `GsimOT.opNorm_smul_of_nonneg` Common/OTDistances.lean:94

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App. A	14–15	"All minima … are attained" for $\delta^{\mathrm{OT}}_{p\to q}$ and $d^{\mathrm{OT}}_{p\to q}$: lower semicontinuity of $Q \mapsto \|B_{p\to q}(G,H,Q)\|$, compactness	`GsimOT.lowerSemicontinuous_opNorm_Bmat` Common/Attainment.lean:136 `GsimOT.isCompact_setOf_isCoupling` Common/Attainment.lean:168 `GsimOT.exists_deltaOTPQ_eq` Common/Attainment.lean:199 `GsimOT.exists_dOT_eq` Common/Attainment.lean:223

Notes on the formalisation

1. **Statement.** `theorem14` is Theorem 14, $\delta_2 \le \delta_2^{\mathrm{OT}} + 2\sqrt{2\delta_2^{\mathrm{OT}}}$. As in the paper, it follows from the bound $\delta_2 \le \gamma_Q + 4\sqrt{\gamma_Q h}$ at the end of the proof (`theorem15_h`) and $h \le \frac{1}{2}$. `theorem15` combines `theorem15_h` with $\delta_2 \le 1$ (`deltaPQ_two_two_le_one`, via $\|\cdot\|_2 \le \|\cdot\|_F \le n$) into $\delta_2 \le \min\{1, \delta_2^{\mathrm{OT}} + 4\sqrt{h\,\delta_2^{\mathrm{OT}}}\} \le \min\{1, \delta_2^{\mathrm{OT}} + 2\sqrt{2\delta_2^{\mathrm{OT}}}\}$, which implies Theorem 14.

2. **Proof structure.** The formalisation follows App. F.2 step by step: `lemma50` uses Lemma 19 and the bilinear bound `eq40_le` behind Eq. (40) (two test vectors); `eq42` and `lemma51` follow the proof of Lemma 51 (via F_{ij} and the expansion $\sum Q(F-m)^2 = \sum QF^2 - \sum m^2$); Eq. (43) is `eq43` (expectation of the squared spectral norm, bounded by the Frobenius norm computed in the eigenbasis, `frobenius_eq_sum_eigvec`); the chain (a) uses Eq. (28) for A'_t (`eq28`) and Eq. (41) (`eq41`); and the proof starts from an optimal Q (`exists_dOT_eq`).

3. **No symmetry reduction.** Instead of assuming $h = \sqrt{p_G(1-p_G)}$ by symmetry (l. 1529), both cases are proved: centre by p_G and use $\|A'\|_F$ (`deltaPQ_le_threshold_G`), or centre by p_H and use $\|B'\|_F$, applying the argument to $(B', A', Q^\top)$ (`deltaPQ_le_threshold_H`). The minimum is then taken.

4. **Normalisations.** The Lean works with $\|B_{2\to 2}(G,H,Q)\|_2 = n\gamma_Q$ (`discKernel`) and with the threshold $s = nt$; all statements are the paper's after division by n. Expectations over the random permutation are weighted sums over a Birkhoff decomposition (`exists_birkhoff_decomposition`).

5. $\delta_2^{\mathrm{OT}} \le 1$ (used after Theorem 14, l. 399) is proved as `deltaOTPQ_two_two_le_one`: $\|B_{2\to 2}(G,H,Q^*)\|_2 \le \|B_{2\to 2}(G,H,Q^*)\|_F \le \sum Q^* \le 1$.